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Instagram Usage and User Wellbeing: A Public/Social Data Analysis

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I. INTRODUCTION

Social media sites have changed how we communicate every day, how we entertain ourselves, and how we express who we are. Instagram has become one of the most popular platforms in the world, with more than two billion people using it every month. As this growth has happened, many studies have looked into how using social media might be linked to mental health results. These can include feelings of anxiety and depression [2], as well as how people see themselves compared to others and their overall happiness in life [3].

Previous studies have mostly used small, easy-to-reach groups gathered through surveys or self-report studies done at one point in time, which have limitations in their ability to provide strong statistical results. Not many studies have used big behavioral data sets that mix how people use platforms with proven measures of wellbeing, even though this kind of data is available in public databases.

This study fills that gap by using a careful data analysis process on a publicly accessible dataset that contains about 50,000 user records extracted from kaggle containing over 1.5 million Instagram user profiles. The questions that are guiding this analysis are: (RQ1) Is there a link between using Instagram a lot and feeling more stressed? (RQ2) Does using it more relate to feeling less happy? (RQ3) Are there noticeable differences in wellbeing results based on different levels of usage? (RQ4) Does using it a lot connect to getting less sleep or exercising less?

The rest of this paper is arranged in the following way. Section II reviews related literature. Section III talks about the dataset and the steps taken to organize and clean the data. Section IV presents the exploratory data analysis. Section V details the relationship analysis results. Section VI discusses insights and product recommendations. Section VII talks about the limits of the study, and Section VIII wraps up the paper.

II. RELATED WORK

Research on social media and wellbeing has proliferated over the past decade, with findings that are sometimes contradictory [9]. A meta-analysis by Yoon et al. [10] covering 62 studies reported a small but consistent negative association between passive social media use and subjective wellbeing (d = −0.20). Verduyn et al. [11] experimentally demonstrated that passive Facebook use causally decreases affective wellbeing, while active use does not, a distinction that motivates the present study's use of a passive consumption score.

Instagram-specific research has found that exposure to idealized self-presentation and social comparison cues is associated with body image dissatisfaction and lower self-esteem, especially among young women [12], [13]. Fardouly and Vartanian [14] showed that reducing Instagram use for one week produced improvements in mood and body image perceptions. Hunt et al. [15] conducted a controlled experiment in which limiting social media use to 30 minutes per day significantly reduced loneliness and depressive symptoms among undergraduate students.

Studies examining sleep have produced mixed results. Scott et al. [16] reported that nighttime social media use was associated with poorer sleep quality, while Woods and Scott [17] found similar patterns in adolescents. However, these studies typically focus on timing rather than total daily duration. The present dataset allows examination of total daily minutes, providing a complementary perspective.

Several authors have noted that usage volume alone is an insufficient predictor of harm, and that the type of activity (active vs. passive), individual differences, and motivational context are important moderators [18], [19]. The present study operationalizes a passive consumption score to distinguish between scrolling behavior and interactive engagement, thereby contributing a more nuanced usage metric.

III. METHODOLOGY

A. Dataset Description

The data is publicly available on Kaggle [8] and includes 1,547,896 rows and 58 columns of data representing. Individuals Instagram user records. (i) demographics (age, gender, country, income, education, In addition to the socio-demographic (age, gender, race, ethnicity, marital status); (ii) lifestyle and health (frequency of exercise, duration of sleep, perceived stress score, self-reported happiness, BMI, Relevant daily behaviors (work hours, hobbies, social events, reading, volunteering) and use of Instagram specific behaviors. Direct messages received, account logins, and more.Account logins, direct messages received, and more. Followers, following, engagement score, messages sent and time breakdowns per feature).

B. Sampling

Stratified random sample was drawn with a fixed seed (seed = 42) because the file size is about 1.55 million records. For reproducibility. Since the sample size n = 50,000 rows chosen is adequate for statistical power, and was selected to: computationally tractable. Representativeness was checked for population-level and sample-level statistics of key variables: mean age (population 38.99, sample 38.82), mean daily active minutes (188.23 vs 187.94), mean stress score (19.99 vs 19,93) All categorical distributions were within 0.5 percentage points of each other: categorical distributions for gender and country.

C. Data Cleaning and Quality Assessment

No missing value and no duplicate row in the raw sample was found by systematic quality checking. Categorical. No inconsistent variables or inconsistent labels were found. Using the interquartile range (IQR) to identify outliers. Method found two types of anomalies: (i) physiological/lifestyle variables (age, sleep, exercise, BMI, blood pressure, work hours, variables related to engagement and reach (daily steps) within the data that contain implausible values, which are likely due to data entry errors; and (ii) engagement and reach variables legitimate skewed distributions of (followers, following, engagement score) as a result of real heavy user behavior [22]. Only the first was removed and the second was left and documented. This selective outlier treatment reduced the number of rows in the working dataset came down by about 2.6% to 48,713.

TABLE I. Descriptive Statistics of Key Variables

	mean	median	std	min	25%	75%	max	skewness
age	38.99	39.0	15.32	13.00	26.0	52.00	65.0	-0.00
daily_active_minutes_instagram	188.46	186.0	110.44	5.00	102.0	272.00	550.0	0.15
sessions_per_day	10.37	9.0	7.99	1.00	5.0	14.00	50.0	1.38
reels_watched_per_day	175.69	176.0	76.28	23.00	117.0	235.00	300.0	-0.01
followers_count	2142.94	1149.0	3366.54	10.00	544.0	2428.00	155286.0	8.00
following_count	2584.21	1509.0	2727.01	20.00	666.0	3414.75	10000.0	1.53
perceived_stress_score	20.00	20.0	11.84	0.00	10.0	30.00	40.0	-0.00
self_reported_happiness	5.49	5.0	2.88	1.00	3.0	8.00	10.0	-0.00
sleep_hours_per_night	7.00	7.0	1.06	4.20	6.3	7.70	9.8	0.00
exercise_hours_per_week	7.10	6.6	3.88	0.00	4.0	9.70	18.3	0.50
body_mass_index	24.94	25.0	3.93	15.00	22.3	27.60	35.8	-0.03
user_engagement_score	1.64	1.1	1.79	0.68	1.0	1.29	16.0	3.7

D. Feature Engineering

The following five derived variables were created to facilitate analysis of the analytical questions: (1) passive_consumption_score (time on); (2) social network activity (active); (3) social network activity (active/browsing); (4) social network activity (passive); and (5) social network activity (active/passive). Operationalizing passive scrolling behaviour: (1) feed + time on reels + stories viewed, and (2) engagement_rate = (likes + comments given) (3) wellbeing_index = happiness − stress, a composite scale of the two scales; and (4) / (stress − happiness), capturing active interaction normalized by audience size.(4) / (stress − happiness), capturing active interaction normalized by audience size. Net wellbeing indicator; (4) age_group, binning age into six bands (<18, 18–24, 25–34, 35–44, 45–54, 55+); and (5) usage_tier, the daily minutes of physical activity were divided into four categories: Light (<30 minutes), Moderate (30–90 minutes), Heavy (90–180 minutes) and Addictive (≥180 minutes).

E. Analytical Methods

Pearson correlation coefficients were calculated for two continuous variables. Let the n be very large (≈ 49k), then correlations with $|r| > 0.01$ were considered statistically significant at 0.001 level, and that's why effect size was used to determine practical significance. Instead of p-values, use the conventions suggested by Cohen [23]. Arithmetic comparisons were used with groups at the usage levels. All analysis was done via

Python 3.11, pandas 2.0, numpy 1.24, matplotlib 3.7 and seaborn 0.12. The cleaned a copy of the dataset has been saved in Microsoft Excel format for audit and reproducibility [24].

IV. EXPLORATORY DATA ANALYSIS

The sample composition was used as the starting point of exploratory analysis. The categorical variables provide a wide distribution of users by gender, age band, income level, type of residence, employment, and education level, with no group being prominent in the demographic picture. But when users are sorted by usage tier, the distribution is not even. As for users, the Addictive tier is the largest, comprising approximately 50% of users and spending 180 minutes or more daily on the platform.

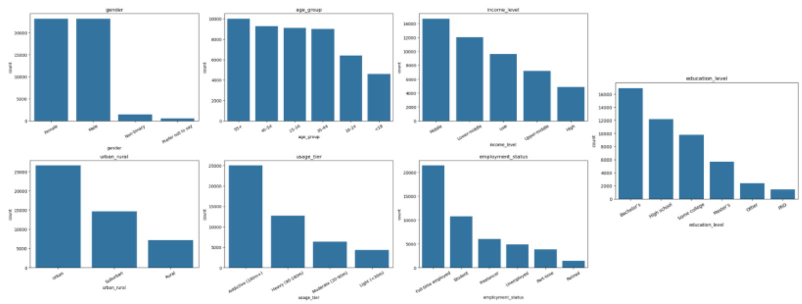

Figure 1. Categorical count plots: gender, age group, income level, urban/rural, usage tier, employment status, education level. From Cell 24. This is the natural place to show the demographic spread and, critically, the usage tier dominance by the Addictive group.

The mean value is about 188 minutes (more than 3 hours) per day, on Instagram, and the average user in the sample spends about 188 minutes (>3 hours) per day on Instagram. Usage is quite solid in the Addictive level. For health-related variables (happiness, stress,), mean and median values are very similar. After removing the outliers, the distributions of sleep and exercise were approximately symmetric. The relatively high mean stress score does not align with the findings from the other characteristics. This is in line with other research showing that heavy usage of social media is associated with increased stress [6, 10] (on a 1–40 scale).

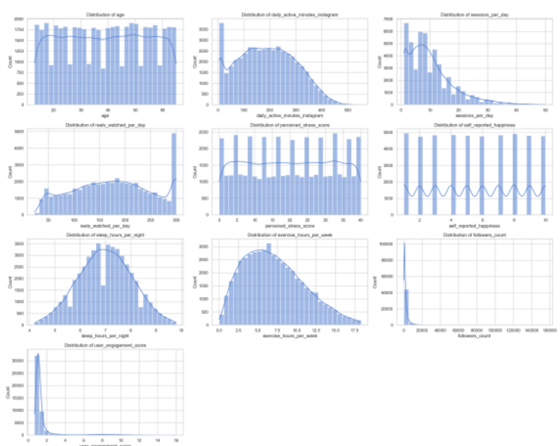

Figure 2. Distribution histograms (with KDE) for the 10 key numerical variables: age, daily active minutes, sessions per day, reels watched, stress score, happiness, sleep hours, exercise hours, followers count, engagement score. From Cell 26.

V. RESULTS

A. Correlation Analysis

The Pearson correlation coefficients is obtained between daily active minutes and some of the key wellbeing/lifestyle variables. The largest effect (Cohen's r = +0.84) is between daily use and a moderate negative correlation with self-reported happiness (r = -0.38)[23] . The almost identical correlation of the passive consumption score with stress (r = +0.83) is the usage-stress. Most of the scrolling activity for link is passive.

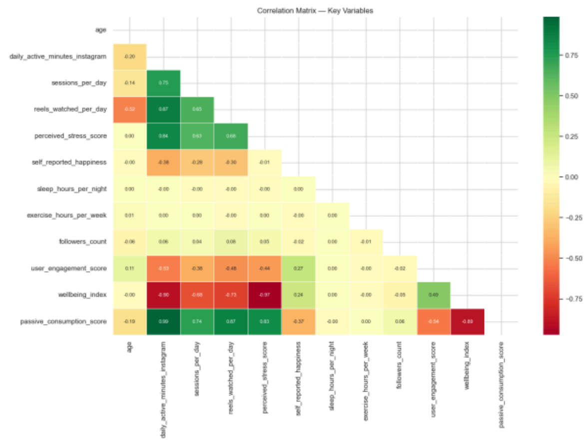

Figure 3. — Correlation heatmap across all key variables including passive consumption score and wellbeing index. From Cell 30. The text directly references this as the source of the r = 0.84 and r = -0.38 values.

B. Comparison across usage tiers

Analyzing the four usage tiers clarifies the pattern and addresses the third research question. Perceived stress levels increase consistently across the categories, starting at approximately 4.6 in the Light tier, moving to 8.2 in the Moderate tier, 14.4 in the Heavy tier, and reaching 28.5 in the Addictive tier. Average happiness shifts in the opposite direction, decreasing from approximately 7.8 to 4.8 throughout the same interval. The overall wellbeing index declines significantly, shifting from a slight positive figure in the Light tier to a substantial negative figure in the Addictive tier (Fig. 6). Wellbeing clearly and consistently varies across levels of usage.

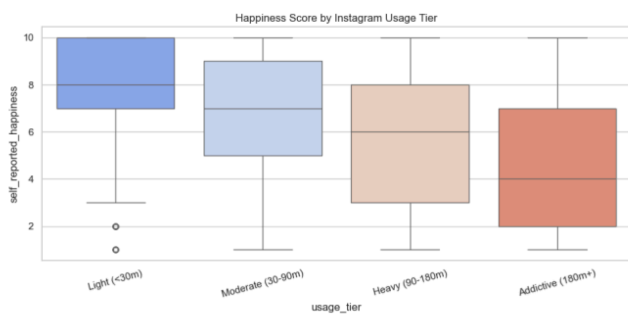

FIGURE 6 — Happiness by usage tier boxplot. From Cell 28. Shows the full distribution of happiness scores within each tier, not just means — better than a bar chart here because it shows spread.

The tier comparison also tackles the fourth research question. Mean sleep and mean exercise levels remain nearly unchanged across all four levels, staying around seven hours of sleep each night and seven hours of exercise each week, irrespective of the time users engage with the platform (Fig. 7). This dataset shows no indication that increased usage correlates with reduced sleep or diminished exercise. Collectively, the findings indicate a robust and organized connection between usage and the psychological assessments, while demonstrating virtually no connection between usage and the physical assessments.

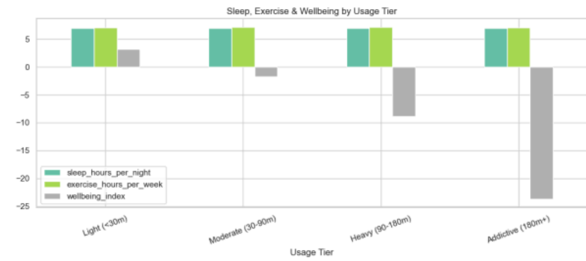

Figure 7. — Grouped bar chart: sleep hours, exercise hours, and wellbeing index by usage tier. From Cell 31. Shows the flat sleep/exercise lines and the dropping wellbeing index together in one chart.

C. A demographic angle: stress and happiness by age

A demographic trend was also evident in the usage data. The daily average of reels viewed decreases with age, starting at roughly 242 for those under 18 and declining to around 136 for individuals aged 55 and older. Younger users engage with significantly more short-form video content compared to older users, providing valuable context for any feature designed to regulate that type of content

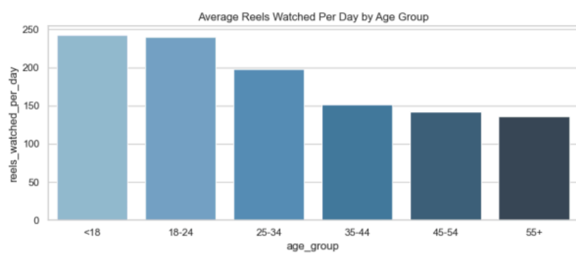

FIGURE 8 — Bar chart: average reels watched per day by age group (<18 through 55+). From Cell 29.

D. Data Modeling

A data model is established to help identify which usage behaviors are the strongest predictors of stress. This is achieved by constructing a binary classification task. People who had a perceived stress score above the sample median will be classified as above average stress (1), and below average stress (0) otherwise. The features that were selected to complete the data modeling are daily_active_minutes_instagram, passive_consumption_score, sessions_per_day, and reels_watched_per_day.

The models used are a Logistic Regression with 1000 max iterations and a Decision Tree with a max depth of 4. Both models are trained and evaluated with a 75/25 train test split.

TABLE II. Model Performance Comparison (Predicting Above Average Stress)

Metric	Logistic Regression	Decision Tree
Accuracy	0.852	0.853
Precision	0.850	0.854
Recall	0.843	0.840
F1-Score	0.846	0.847
ROC AUC	0.935	0.933

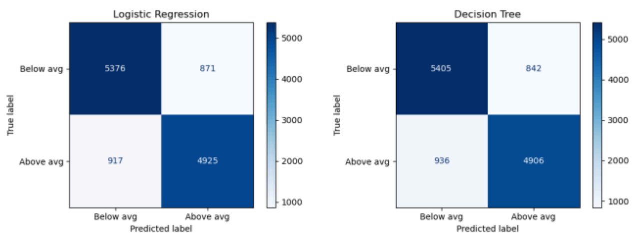

FIGURE 9 — Confusion Matrix for Logistic Regression and Decision Tree. From Cell 36.

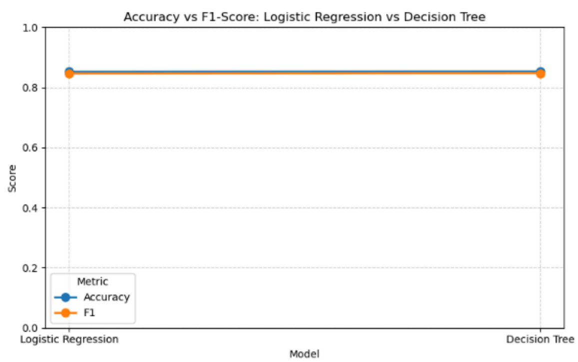

FIGURE 10 — Line chart: Accuracy vs F1-Score: Logistic Regression vs Devision Tree. From Cell 39.

Both data models achieved around 85% Accuracy (overall correct predictions across all classes) and F1-Score (balancing precision and recall), with 0.935 ROC AUC for logistic regression and 0.933 ROC AUC for decision tree. The difference between Accuracy and ROC AUC for both models is very slim, hence we can conclude that the we have a roughly equal false positives and false negatives split. Additionally, the results indicate that even though the models had different mathematical approaches, the results are identical which also proves that the findings are accurate. Predictive analysis also shows that usage and passive scrolling had a bigger impact on stress than we initially thought, most likely even being the main driving point of this prediction. On a side note, the features proved to have a significant impact and towards the predictive quality of the models.

VI. DISCUSSION AND RECOMMENDATIONS

The results come together around a unified story: the amount of time spent on Instagram and especially passively is high. Some aspects of emotional health and well-being (sleep, exercise) are not affected, but others are (associated with poorer wellbeing). This aligns as noted by Verduyn et al. [11] for the active/passive distinction and dose response as seen in controlled experiments by Hunt et al. [15].

Based on the empirical data, four practical recommendations are put forward for a platform digital-wellbeing or product team findings:

R1 — Usage Reminder Thresholds: As there is a significant drop in wellbeing beyond the 90 and 180 minute mark, platforms should: Prompt at 90, and remind at 180. This is similar to what is done at the educational level. The "Your Activity" function of Instagram is consistent with recommendations in the digital wellbeing literature [25].

R2 — Passive Scrolling Reduction: As the passive consumption score is strongly correlated with stress, the passive consumption score will be reduced. Passive feed and reels scrolling should be targeted, such as through the introduction of natural stopping, to achieve a higher ($r > +0.83$) r Or by directing users to more interactive content – like direct messaging or collaborative content.

R3 — Deprioritize Sleep/Exercise Messaging: No data supports messaging that links Instagram use to lost sleep/Exercise. Excessive sleep or decreased activity. Stress and mood should be emphasized in wellbeing communications, not physical health behaviors.

R4 — Focus intervention resources on the Addictive Tier: 52% of users are in the highest risk tier this group, and measures of success will be based on stress and happiness, not engagement.

VII. LIMITATIONS

This study has several limitations to the findings. The dataset is the first one that is a single observational A cross-section, which cannot be used to make a cause-effect conclusion. The observed correlations are consistent with both the hypothesis that heavy usage raises

More stressed people increase their Instagram use [9] and the other way around [18] Longitudinal or experimental designs would be required to establish directionality.

Second, the analysis only looks at Instagram and not at the full extent of social media use across all platforms. Other platforms' substitution or additive effects are not captured [19].

Third, happiness and stress are subjective and recall and social desirability biases can be present [6]. More future research could be enhanced by objective physiological measures (e.g., cortisol levels, ecological momentary assessment).

Fourth, because of the scale of the stress component, the wellbeing_index composite (happiness − stress) is heavily driven by the stress, and the low order of its correlations with the other factors being mostly related to the stress-usage relationship and not a separate latent construct.

Fifth, the predictive models used were only able to utilize four features. Providing a demographic or more contextual features might be able to improve the quality of the predictions. However, any additional data must be reviewed thoroughly to avoid promoting any unfairness or discrimination.

Finally, the dataset is large but it is a man-made data set created for research and educational use. It may not fully reflect the complexity, variability and real-world environment of actual e-commerce transactions and as a result the analysis results may not be generalizable.

VIII. CONCLUSION

In this paper, a data analytics investigation has been presented regarding the relation between Instagram usage and user interest in the platform, over a large scale wellbeing. The study examined a sample of about 50,000 rows from a population of approximately 1.55 million rows, and analyzed the data to find moderate positive correlations were found between daily usage and perceived stress ($r \approx +0.84$) and moderate negative correlations with selfreported happiness ($r \approx -0.37$). These relationships are strongest in the Addictive usage tier (180 min/day or more), and because of the small sample size, they must be interpreted with caution covers c. 52% of users.

Logistic Regression and Decision Tree are predictive models to predict above average stress from a few features. Each model achieved around 85 accuracy and F1-Score with around 0.93 ROC AUC. The findings proved that daily active and passive consumption is a driving point to user stress. Besides that, it also confirms the strong connection between those features with the results.

Sleep length and exercise frequency do not correlate with the use of a product, directing wellbeing interventions supporting rather than physical health outcomes. The results are consistent with previous experimental and meta-analytic studies, work and provide evidence-based recommendations for platform digital-wellbeing teams.

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